

Smart Microgrid Energy Management System
Using Deep Reinforcement Learning

Submitted in partial fulfillment of the requirements for the degree of

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by

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DECLARATION

I hereby declare that the project report entitled "**Smart Microgrid Energy Management System Using Deep Reinforcement Learning**", submitted by me for the award of the degree of Bachelor of Technology in Electrical and Electronics Engineering to VIT University, is a record of bonafide work carried out by me under the supervision of Dr. Janaki M, Assistant Professor, School of Electrical Engineering, VIT University, Vellore.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Executive Summary

Traditional microgrids fail to economically manage the stochastic intermittency of solar generation and dynamic Time-of-Use (ToU) grid tariffs because standard rule-based heuristic controllers lack predictive foresight regarding lagging thermal HVAC loads. This thesis directly addresses this inefficiency by architecting a dual-layer Artificial Intelligence Energy Management System for decentralized microgrids.

To accomplish this, a Long Short-Term Memory (LSTM) sequence-to-sequence neural network was engineered using a 5-year, 52,608-hour NASA meteorological dataset to accurately forecast 24-hour localized array outputs. Subsequently, a Proximal Policy Optimization (PPO) Deep Reinforcement Learning agent was structured upon a continuous Markov Decision Process. This agent actively controls a simulated 180 kWh industrial Battery Energy Storage System (BESS). A Glassmorphic SCADA-style dashboard was implemented to provide "Explainable AI" parameter transparency alongside IEEE-standard hardware overrides.

Findings indicate the PPO algorithm successfully converged across 20,000 tensor timesteps. It autonomously executed extreme energy arbitrage — pre-charging at ₹3/kWh and discharging to avoid ₹11.5/kWh peaks — completely safely within the strict 20%–95% battery limits. The architecture reliably reduced operational expenditures by 12.23% against deterministic baselines, successfully insulating the microgrid operating budget even during an unpredictable +24% facility load surge.

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LIST OF TERMS AND ABBREVIATIONS

AdamW	— Adaptive Moment Estimation with Weight Decay (optimizer)
BESS	— Battery Energy Storage System
CSV	— Comma-Separated Values
CSS	— Cascading Style Sheets
CUDA	— Compute Unified Device Architecture (NVIDIA)
DER	— Distributed Energy Resource
DNI	— Direct Normal Irradiance
DQN	— Deep Q-Network
DRL	— Deep Reinforcement Learning
DSS	— Decision Support System
EDA	— Exploratory Data Analysis
EMS	— Energy Management System
FastAPI	— Fast Application Programming Interface (Python web framework)
GAE	— Generalized Advantage Estimation
GHI	— Global Horizontal Irradiance
GPU	— Graphics Processing Unit
HCI	— Human-Computer Interaction
HMI	— Human-Machine Interface
HVAC	— Heating, Ventilation, and Air Conditioning
IEC	— International Electrotechnical Commission
IEEE	— Institute of Electrical and Electronics Engineers
IoT	— Internet of Things
JSON	— JavaScript Object Notation
kWh	— Kilowatt-hour
LSTM	— Long Short-Term Memory
MAE	— Mean Absolute Error
MARL	— Multi-Agent Reinforcement Learning
MDP	— Markov Decision Process

MILP — Mixed Integer Linear Programming
MSE — Mean Squared Error
NASA — National Aeronautics and Space Administration
OPEX — Operational Expenditure
P2P — Peer-to-Peer
PPO — Proximal Policy Optimization
PV — Photovoltaic
REST — Representational State Transfer
RL — Reinforcement Learning
ROI — Return on Investment
SCADA — Supervisory Control and Data Acquisition
SoC — State of Charge
SVG — Scalable Vector Graphics
ToU — Time-of-Use
XAI — Explainable Artificial Intelligence

CHAPTER 1

INTRODUCTION

1.1 Objective

The primary objective of this research project is to architect, synthesize, and rigorously validate an intelligent, Edge-Compute capable Decision Support System (DSS) purpose-built for decentralized microgrid environments. The modern electrical grid is undergoing a fundamental structural transformation: centralized fossil-fuel generation points are progressively being displaced by geographically distributed, localized energy resources (DERs) — solar photovoltaic arrays, wind turbines, and Battery Energy Storage Systems — that are controlled at the facility level rather than at the utility level. Successfully operating these microgrids at optimal economic efficiency demands millisecond-scale decision-making to exploit dynamic Time-of-Use (ToU) energy arbitrage: acquiring inexpensive grid energy during off-peak tariff windows, storing it within battery reserves, and deploying it precisely when macrogrid import tariffs escalate to peak rates. The central technical objective of this system is to mathematically minimize microgrid Operational Expenditure (OPEX) by replacing rigid, pre-programmed heuristic dispatch algorithms with an autonomous Deep Reinforcement Learning (DRL) agent trained via the Proximal Policy Optimization (PPO) algorithm, capable of predicting and acting upon shifting climatic and consumption variables in real time.

A secondary — yet equally critical — objective of this project is to resolve the widely documented 'Black Box' transparency problem that has historically impeded the deployment of Artificial Intelligence in industrial IoT and power infrastructure sectors. Grid operators and facility managers remain hesitant to grant autonomous control authority to neural networks because the internal decision matrices of these models are opaque and unauditable. To address this, an explicit design objective was to engineer a professional-grade, industrial Human-Machine Interface (HMI) modeled visually and functionally after enterprise SCADA systems. This interface natively translates the

continuously fluctuating mathematical action-space outputs of the DRL agent into deterministic, human-readable Explainable AI (XAI) reasoning strings, provides physical SCADA-style override controls, and enforces absolute hardware safety limits independent of the AI layer.

1.2 Motivation

The compelling global imperative to transition toward sustainable, renewable-dominant energy paradigms is fundamentally constrained by the severe stochastic intermittency inherent to solar photovoltaic and wind generation. Microgrids offer transformative potential for localized infrastructure — hospital campuses, heavy industrial manufacturing plants, university facilities, and smart city blocks — to operate with significant independence from the primary macrogrid, shielding themselves from rolling blackouts and exorbitant dynamic tariff surges. However, realizing this potential requires energy management intelligence that dramatically exceeds what conventional rule-based controllers can provide.

Contemporary threshold-based energy controllers are fundamentally inadequate for this task. Hard-coded logical constructs of the form 'if battery SoC falls below 40%, import from grid' lack any capacity for forward prediction or temporal reasoning. They are operationally blind to the future. Critically, they cannot proactively recognize that a sustained high ambient temperature at 14:00 will inevitably produce a massive HVAC electrical demand surge two to three hours later at approximately 16:30 — a lagging thermal correlation confirmed empirically in the EDA phase of this project. This temporal blindness causes reactive energy purchasing at the economically worst possible moments, nullifying the financial justification for owning expensive Battery Energy Storage Systems and rendering the investment in renewable generation infrastructure suboptimal.

This research was motivated by the desire to bridge the wide chasm separating advanced academic Deep Learning theory from tangible, industrial-grade deployment.

By fusing a Long Short-Term Memory (LSTM) sequence-to-sequence network — capable of modeling the complex, lagged relationships between meteorological variables and facility load demand — with a PPO agent capable of acting upon that forecast through optimal battery dispatch, this project demonstrates that AI-driven EMS architectures can be both economically superior to legacy approaches and sufficiently transparent to earn operator trust. The project further demonstrates that sophisticated machine learning systems can be packaged within production-quality, real-time monitoring interfaces rather than confined to offline research notebooks.

1.3 Background

A microgrid is a locally bounded, controllable network of distributed electrical energy generators, flexible loads, and storage assets that can operate either in grid-connected mode — synchronized with and drawing from the central macrogrid — or in islanded mode, functioning as an autonomous power island disconnected from the wider grid. The Microgrid Energy Management System (EMS) serves as the central computational intelligence governing where power flows within this bounded network at every operational timestep.

Historically, microgrid EMS architectures employed deterministic mathematical optimization frameworks, most prominently Mixed Integer Linear Programming (MILP). A MILP solver accepts a fixed system model, a weather forecast, and a tariff schedule as inputs, and produces a globally optimal dispatch schedule for the upcoming operational period. While mathematically proven to yield optimal solutions within their formulation assumptions, MILP approaches carry fundamental limitations that render them unsuitable for modern volatile microgrid environments. First, their computational complexity scales exponentially with problem size — recalculating an optimized dispatch route to respond to a sudden cloud transient that severs 80% of solar generating capacity within a single minute requires seconds to minutes of solver execution time, creating catastrophic lag during which the grid destabilizes. Second, MILP assumes perfect forecast accuracy, an assumption that is physically unrealistic in meteorologically complex environments.

Recent advances in Machine Learning — specifically Reinforcement Learning (RL), which is structurally formalized through the continuous Markov Decision Process (MDP) framework — offer a paradigm-shifting alternative. An RL agent enters a simulated environment without prior knowledge of its governing physics, interacts with it over thousands of training episodes, and autonomously deduces an optimal control policy through cumulative reward maximization. Once training reaches convergence, executing the learned policy requires only a forward pass through the policy network — computationally completing in under one millisecond — granting the EMS the ability to issue instantaneous, near-optimal dispatch commands that respond to real-time state changes without any computational lag. Proximal Policy Optimization (PPO), introduced by Schulman et al. (2017), represents the current state-of-the-art policy gradient algorithm for continuous action spaces of the type required for precise kilowatt-level battery dispatch control.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 Review of Literature

The field of AI-driven microgrid energy management has evolved rapidly over the past decade, progressing from tabular reinforcement learning on simplified grid topologies to deep neural network policies operating over high-dimensional continuous state spaces. The following review surveys the foundational and contemporary works that directly informed the architectural decisions made in this project.

Kuznetsova et al. (2013) were among the first to systematically apply reinforcement learning to microgrid energy dispatch. Working with a grid incorporating solar generation and a single battery storage unit, they formulated the dispatch scheduling problem as a finite Markov Decision Process and applied standard tabular Q-learning. Their findings established that an RL agent could learn cost-reducing dispatch policies purely through trial-and-error interaction with a simulated environment, without requiring an explicit mathematical model of the optimization problem. However, their tabular approach required discretizing the battery state-of-charge into a small number of coarse bins, introducing systematic suboptimality and scaling poorly to systems with high-dimensional or continuous state spaces — a limitation directly addressed in the present work through the adoption of deep neural network function approximators.

Francois-Lavet et al. (2016) extended Q-learning to the deep neural network domain, applying Deep Q-Networks (DQN) to microgrid scheduling and demonstrating substantial cost improvements over heuristic benchmarks. Their work established that deep neural networks could successfully approximate the Q-function over large, complex state spaces relevant to real energy management problems, and confirmed the importance of experience replay buffers and target network stabilization in achieving convergence. These foundational techniques are embedded in the Stable Baselines3 PPO implementation used in the present project.

Kofinas, Dounis and Vouros (2018) investigated a multi-agent decentralized architecture in which individual Fuzzy Q-learning agents were assigned to separate microgrids within a networked topology, sharing aggregate demand information to coordinate energy flows across the network. Their use of fuzzy membership functions to represent continuous battery states improved upon coarse tabular discretization, though the fuzzy Q-learning approach still required manual specification of the membership function boundaries and could not learn arbitrary non-linear policies from raw data. The multi-agent concept introduced in their work directly informs the future scope of the present project, which envisions extension to Multi-Agent Reinforcement Learning (MARL) across smart city microgrid clusters.

Mnih et al. (2015) established the foundational methodology for human-level control through deep reinforcement learning using the DQN algorithm, demonstrating super-human performance across 49 Atari game environments from raw pixel inputs. While not a power systems paper, this seminal work introduced experience replay, convolutional feature extraction, and target network weight freezing as standard practice in DRL — all of which are incorporated in the PPO implementation used in this project. The work's demonstration that a single algorithm with fixed hyperparameters could achieve expert-level performance across diverse environments provided the theoretical basis for applying DRL to the microgrid dispatch domain.

Venayagamoorthy et al. (2016) proposed a dynamic energy management system for smart microgrids employing echo state networks for real-time load and generation forecasting, coupled with a neural network controller for dispatch optimization. A key contribution of this work was the explicit modeling of battery degradation as a time-varying cost within the control objective, recognizing that repeated deep charge-discharge cycles accelerate lithium-ion cell capacity fade. While the present project does not model cycle-level degradation, this consideration — enforcing hard SoC boundaries of 20%–95% to prevent the most damaging operating modes — is directly incorporated into the reward penalty mechanism and edge override layer.

Rocchetta et al. (2019) applied a reinforcement learning framework to the joint problem of optimal grid operation and asset maintenance scheduling. Their treatment of safety constraints within the RL reward signal — specifically the use of large numerical penalties for policy actions that violated physical system limits — directly informs the constraint-enforcement mechanism in the present work's custom Gym environment, where a penalty of -10,000 is levied for any action that would breach the battery's electrochemical safety boundaries.

Ji et al. (2019) conducted a real-time energy management study for a microgrid using deep reinforcement learning, comparing DQN and Double DQN variants across multiple renewable penetration scenarios. Their results confirmed that DRL agents adapted effectively to varying renewable availability conditions and outperformed Model Predictive Control (MPC) benchmarks under high meteorological uncertainty. Importantly, Ji et al. reported that the inference speed of a trained DRL policy — sub-millisecond per decision — offered a decisive practical advantage over MPC's iterative solving procedure, a finding that validates the algorithmic selection made in this project.

Singh et al. (2024) demonstrated machine learning-based energy management and power forecasting in grid-connected microgrids with multiple distributed energy sources, achieving an MSE of 2.002 for solar PV and 3.059 for wind power using Support Vector Regression. Their study confirmed that ML-based forecasting significantly outperforms traditional linear regression across all accuracy metrics, and that combining accurate forecasting with intelligent storage dispatch produces measurable reductions in peak grid import. The SVR results provide a useful external benchmark against which the LSTM MAE values achieved in the present project (2.0748 kW solar, 3.5385 kW load) can be contextualized.

Spiliopoulos et al. (2025) presented a Smart Microgrid Platform integrating AI and Deep Reinforcement Learning for sustainable energy management, reporting a 23%

reduction in energy costs, a 40% decrease in grid dependency, and 85% renewable energy utilization. Their study employed PPO agents and found PPO to be up to 9.2% more effective in reducing annual costs compared to Rule-Based Controllers, with PPO's clipped surrogate objective identified as the primary factor in its superior training stability — a finding that directly corroborates the algorithm selection rationale in the present project.

Dusabimana and Yoon (2020) provided a comprehensive taxonomic survey of microgrid EMS methodologies, categorizing approaches into rule-based, optimization-based, and AI-based families and systematically comparing their performance characteristics. Their analysis explicitly identifies Deep RL as the most promising direction for future EMS development, particularly for environments where the operational state space is large, the reward signal is temporally delayed, and real-time responsiveness is non-negotiable — precisely the conditions characterizing the ToU-arbitrage problem addressed in this thesis. The survey's identification of operator transparency as a critical barrier to industrial AI adoption directly motivated the XAI-integrated dashboard design that constitutes a central contribution of this project.

2.2 Project Description

This project constitutes an exhaustive, end-to-end, full-stack intelligent Energy Management System for a simulated high-demand industrial microgrid facility. The architecture is explicitly structured into two decoupled processing environments: a Heavy-Compute AI Training and Inference Backend, and a Lightweight Edge-Compute Presentation Frontend — a separation mandated by the computational mismatch between the deep learning training phase and the real-time operational environment.

The predictive intelligence layer employs a Long Short-Term Memory (LSTM) sequence-to-sequence neural network trained on 52,608 hours of longitudinal meteorological observations sourced via the NASA POWER Application Programming Interface. The dataset spans five complete calendar years of hourly readings, capturing

the full spectrum of seasonal, diurnal, and weather-event-driven variability in solar irradiance, ambient temperature, relative humidity, and wind speed. Two independent LSTM models were constructed: one forecasting solar generation output in kilowatts, and one forecasting facility electrical load demand. Both models incorporate advanced temporal feature engineering — including sine-cosine harmonic encoding of the hour-of-day to eliminate the discontinuity at midnight transitions — and rolling statistical features derived from three-hour and twenty-four-hour lag windows. The normalized 48-step input sequences allow the LSTM's gated recurrent cells to capture the multi-hour temporal dependencies that characterize lagging HVAC load responses to ambient temperature fluctuations.

The prescriptive intelligence layer is a Deep Reinforcement Learning agent based on the Proximal Policy Optimization (PPO) algorithm, trained within a custom OpenAI Gym Python environment designated MicrogridPPOEnv. This environment mathematically enforces the electrochemical physics of an industrial 180 kWh Lithium-Ion Battery Energy Storage System, including capacity limits, maximum charge and discharge power rates, round-trip efficiency losses, and hard SoC boundaries. At each hourly timestep, the environment presents the agent with a five-dimensional state vector comprising the current solar output, wind generation, facility demand, battery SoC, and applicable grid tariff. The agent responds with a continuous action scalar mapping the desired battery power flow, and receives a scalar reward encoding the economic cost of the resulting grid import. Over 20,000 training timesteps, the PPO agent converged on a policy that proactively exploits the ToU tariff structure, pre-charging during the overnight off-peak window (₹3/kWh) and discharging during the evening peak period (₹9–11.5/kWh) to minimize cumulative grid expenditure.

The presentation layer is a high-performance, fully asynchronous React.js 18 web application that functions as a simulated edge-compute node connected to the microgrid switchgear. Operating on a deterministic 1.5-second physics tick driven by React's `useEffect` interval hook, the dashboard provides operators with live power flow telemetry, 12h/24h/48h predictive horizon charts rendered via Recharts SVG, timestamped anomaly alert logs, an Explainable AI banner that converts the agent's

continuous action scalar into readable English dispatch reasoning, and a SCADA-style Operator Override Panel implementing physical Island Mode, Force Charge, and Force Discharge controls that enforce hardware SoC boundaries independently of the AI inference layer. To maintain dashboard operability during backend disconnection events, a native JavaScript fallback simulator utilizing smooth trigonometric noise functions replicates the physics engine locally in the browser.

2.2.1 Software

The entire computational stack is built on open-source, freely available frameworks, eliminating direct software licensing costs for the project. The AI intelligence backbone is implemented in Python 3.10, with the LSTM forecasting models constructed using PyTorch 2.12 — chosen for its dynamic computational graph, efficient CUDA integration, and Pythonic tensor manipulation API. The PPO reinforcement learning agent is implemented using the Stable Baselines3 library (version 2.0+), which provides a standardized, research-validated PPO implementation compatible with custom OpenAI Gym environments. Data engineering — encompassing API ingestion, normalization, feature engineering, and DataFrame manipulation of the 52,608-row NASA dataset — is performed using Pandas and NumPy. The backend inference and API layer is structured as a RESTful FastAPI endpoint, serving JSON payload responses to the frontend polling interval.

The frontend HMI application is built with React.js 18 and compiled via the Vite build tool for optimized bundle performance. Chart rendering employs the Recharts library for efficient SVG-based, animated time-series visualization, while the Lucide-React library provides scalable vector iconography throughout the interface. All visual styling is implemented through bespoke Vanilla CSS3 — eschewing framework overhead in favor of precise, hardware-accelerated backdrop-filter rendering for the glassmorphic panel aesthetics. Version control across the full project lifecycle is managed through Git and the GitHub platform.

2.2.2 Hardware

The LSTM and PPO model training phases were conducted on a consumer-grade x64 workstation configured with a CUDA-enabled NVIDIA RTX series GPU, which provided the parallel tensor computation acceleration necessary for efficient stochastic gradient descent over the large training dataset and the multi-episode RL training loop. The host system maintained a minimum of 16 GB of RAM to support in-memory manipulation of the full 52,608-row Pandas DataFrame without disk I/O bottlenecks during the feature engineering phase.

The Battery Energy Storage System is implemented entirely as a software model. The simulated BESS specifications — 180 kWh nominal capacity, 55 kW maximum charge intake, 58 kW maximum discharge output, and a round-trip electrochemical efficiency coefficient of 0.965 — are encoded as hard physical constraints within the custom Gym environment, ensuring that all AI training and real-time inference operates within bounds that accurately reflect industrial Lithium-Ion hardware characteristics. The solar photovoltaic array is modeled at 55 kW peak generating capacity, scaled dynamically against the true local Global Horizontal Irradiance (GHI) and Direct Normal Irradiance (DNI) coordinates retrieved from the NASA POWER dataset. No dedicated hardware prototype was constructed; the system is validated exclusively through software simulation.

2.3 Goals

The first project goal is to conduct a rigorous Exploratory Data Analysis (EDA) on the five-year, planetary-scale NASA meteorological dataset to identify, isolate, and formally characterize the complex non-linear correlations — including the 2.5-hour lagging thermal indicator between ambient temperature spikes and HVAC load surges — that render rule-based EMS controllers structurally inadequate for modern volatile microgrid environments.

The second goal is to design, mathematically constrain, train, and deploy a Continuous Action Space Proximal Policy Optimization (PPO) DRL agent that demonstrably outperforms all standard heuristic threshold models in the exploitation of Time-of-Use tariff arbitrage, validated by quantitative comparison of grid import costs over identical test periods.

The third goal is to engineer a fully production-ready 'Control Room' edge dashboard capable of rendering high-frequency telemetry updates with a 1.5-second refresh cycle without browser memory leaks or frame drops, providing operators with a continuously accurate view of all critical system parameters across multiple forecast horizons.

The fourth goal is to overcome the primary barrier to industrial AI adoption by implementing an Explainable AI (XAI) module that dynamically maps the DRL agent's raw continuous action scalars to explicit, human-verifiable dispatch reasoning strings displayed on the SCADA interface, thereby providing the transparency and auditability required for operator trust in safety-critical environments.

The fifth and final goal is to validate the safety and robustness of the complete architecture under adversarial conditions — including a simulated +24% HVAC facility load surge and a complete macrogrid disconnection scenario (Island Mode) — confirming that the edge-layer physical constraint enforcement mechanism operates correctly and independently of the AI decision layer under all stress conditions tested.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 Software Specification

The complete software ecosystem spans three functional tiers — Data Engineering, AI Intelligence, and Edge Presentation — each comprising tools selected for technical suitability, open-source availability, and mutual interoperability. Table 3.1 provides the full specification.

Table 3.1 Core Software Ecosystem — Versions and Architectural Roles

Package / Tool	Version	Deployment Tier	Architectural Role
Python	3.10+	AI Core / Backend	Primary language; differential equations, REST API orchestration
PyTorch / CUDA	2.12.0+	AI Intelligence	LSTM sequence-to-sequence neural network construction and GPU-accelerated training
Stable Baselines3	≥ 2.0	AI Intelligence	Standardized, research-validated PPO Reinforcement Learning agent
OpenAI Gym	0.26+	AI Intelligence	Custom MicrogridPPOEnv — physics-bound simulation environment
Pandas / NumPy	Latest	Data Engineering	52,608-row NASA dataset ingestion, normalization, feature engineering
FastAPI	Latest	Backend API	RESTful JSON inference endpoint linking AI outputs to frontend
React.js / Vite	18+	Edge Presentation	Asynchronous component architecture; 1.5s telemetry lifecycle hooks
Recharts	2.0+	Edge Presentation	SVG-based, animated time-series visualization for forecast charts
Lucide-React	Latest	Edge Presentation	Scalable vector iconography throughout the SCADA dashboard
Vanilla CSS3	—	Edge Presentation	Custom glassmorphic styling; hardware-accelerated backdrop-filter rendering
Git / GitHub	—	DevOps	Version control and source tracking throughout project lifecycle

3.2 Hardware Specification

Hardware resources serve two distinct purposes in this project: the physical workstation used for AI model training and inference, and the simulated physical assets that the EMS governs. Table 3.2 details both categories with their operational parameters.

Table 3.2 Hardware Subsystems and Physical Constraint Parameters

Subsystem	Type		Specification / Parameter
Processing Unit	Real	— Training Hardware	x64 Intel Core i7 / AMD Ryzen 7 (minimum 8 cores)
GPU Accelerator	Real	— Training Hardware	NVIDIA CUDA-enabled RTX series (CUDA 11.x or later)
System Memory	Real	— Training Hardware	16 GB+ RAM (for in-memory Pandas DataFrame operations)
BESS Capacity	Simulated	Software in	180 kWh nominal Lithium-Ion aggregate capacity
Charge Inverter	Simulated	Software in	55 kW maximum charge intake (current rectifier limit)
Discharge Inverter	Simulated	Software in	58 kW maximum discharge output (inverter thermal limit)
Round-trip Efficiency	Simulated	Software in	$\eta = 0.965$ per charge / discharge cycle (heat loss modelling)
SoC Operating Range	Simulated	Software in	20.0% floor (cell-death threshold) — 95.0% ceiling (thermal runaway prevention)
Solar PV Array	Simulated	Software in	55 kW peak generating capacity; scaled against NASA GHI/DNI data

3.3 Standards and Prior Work

All physical rule-sets encoded within the OpenAI Gym Python script and the React GUI conform strictly to contemporary microgrid standards established by the Institute of Electrical and Electronics Engineers (IEEE) and the International Electrotechnical Commission (IEC).

IEEE 1547 (Standard for Interconnecting Distributed Resources with Electric Power Systems) governs the technical requirements for connecting distributed energy resources to the electric power system. In this project, the simulated grid import ceiling

and the mandatory load-isolation behaviors activated during Island Mode testing are modeled in direct accordance with the principles established by this standard.

IEEE 2030.7 (Standard for the Specification of Microgrid Controllers) prescribes the functional architecture of microgrid control systems, mandating a structural separation between the dispatch intelligence layer and the human supervisory interface. This project explicitly adheres to this separation: the AI inference engine and the SCADA override panel are implemented as independent software modules communicating through a well-defined RESTful API boundary.

IEC 61850 (Communication Networks and Systems for Power Utility Automation) was used as the conceptual data-modeling reference for the telemetry property mapping between the simulated switchgear states and the central command interface. The hierarchical data model structure — distinguishing logical nodes, data objects, and functional constraint classes — informed the JSON schema architecture of the system's API payload design.

CHAPTER 4

DESIGN APPROACH AND DETAILS

4.1 Design Approach and Mathematical Modelling

The complete system was developed through a structured, six-phase methodology in which each phase built upon verified deliverables from the preceding phase. The design approach begins with planetary-scale data engineering, progresses through the construction of a physics-bound simulation environment, continues through AI model training and convergence validation, proceeds through interface architecture and asynchronous integration, and concludes with robustness validation under adversarial stress conditions.

4.1.1 Phase 1: Data Engineering and Exploratory Data Analysis

The dataset was sourced via the NASA POWER Application Programming Interface, queried against the GPS coordinates of a representative mid-latitude industrial facility to obtain 52,608 hours of hourly meteorological measurements spanning January 2019 to December 2023. Key variables extracted included Global Horizontal Irradiance (GHI, W/m^2), Direct Normal Irradiance (DNI, W/m^2), ambient dry-bulb temperature (T2M, $^{\circ}\text{C}$), relative humidity (RH2M, %), and wind speed at 10m elevation (WS10M, m/s). Facility load demand was derived from the temperature and irradiance inputs through a calibrated physical model.

All input variables were normalized to the $[-1.0, 1.0]$ range using scikit-learn MinMaxScaler to prevent gradient dominance during neural network training — a critical preprocessing step given the orders-of-magnitude difference between raw GHI values (0 to 1000 W/m^2) and load demand values (38 to 120 kW). Temporal features were encoded as continuous cyclic variables using sine-cosine harmonic transformations: $\text{hour_sin} = \sin(2\pi \times \text{hour} / 24)$ and $\text{hour_cos} = \cos(2\pi \times \text{hour} / 24)$. This representation ensures the neural network understands that 23:59 and 00:01 are

temporally proximate — a critical property for models learning diurnal charging patterns — and eliminates the spurious discontinuity at midnight. Rolling statistical features were computed over three-hour and twenty-four-hour sliding windows: the three-hour rolling mean and standard deviation capture short-term generation trend momentum, crucial for detecting the onset of cloud-cover events, while the twenty-four-hour rolling statistics provide awareness of the current day's overall solar resource quality relative to the historical baseline. The final input feature vector thus contains: [GHI, DNI, T2M, RH2M, WS10M, hour_sin, hour_cos, GHI_3h_mean, GHI_3h_std, Load_24h_mean] — a ten-dimensional observation at each hourly timestep, assembled into sequences of 48 consecutive hours before being passed to the LSTM input gate.

Exploratory Data Analysis confirmed three critical findings that shaped the system architecture. First, solar irradiance exhibits extreme bimodal volatility: output frequently collapses from peak generating capacity to near-zero within a single hour due to localized adiabatic cloud formation, rendering smooth parabolic generation assumptions entirely invalid. Second, a statistically significant lagging correlation exists between sustained ambient temperature spikes and HVAC electrical load surges, with a measured time lag of approximately 2.5 hours between the thermal stimulus and the demand response — a relationship that linear regressors cannot capture but LSTM models can naturally represent through their internal hidden state. Third, the facility demand profile exhibits a severe evening peak coinciding exactly with the cessation of solar generation: base overnight load of approximately 38 kW rises to a morning plateau of approximately 60 kW, then surges to in excess of 120 kW between 18:00 and 21:00 — precisely the window when solar output reaches absolute zero at sunset. These three findings collectively establish the empirical necessity of the dual-layer LSTM-DRL architecture adopted in this project.

4.1.2 Phase 2: MDP Formulation and Physics-Bound Environment

The microgrid energy dispatch problem was formally modeled as a continuous-state Markov Decision Process defined by the five-tuple (S, A, P, R, γ) , where S denotes the state space, A the action space, P the state transition probability function, R the reward

function, and γ the temporal discount factor. The state vector presented to the agent at each hourly timestep is defined as:

$$S_t = [P_{solar,t}, P_{wind,t}, P_{load,t}, SoC_t, Tariff_t] \quad \dots (4.1)$$

where $P_{solar,t}$ is the current solar photovoltaic output in kW, $P_{wind,t}$ is the wind generation in kW, $P_{load,t}$ is the facility electrical demand in kW, SoC_t is the battery state-of-charge expressed as a percentage, and $Tariff_t$ is the applicable grid import cost in ₹/kWh at timestep t .

Battery dynamics are governed by electrochemical charge-balance equations that explicitly account for heat-loss inefficiencies. During a charging operation (net battery power $P_{bat} < 0$, i.e., power flowing into the battery):

$$SoC_{(t+1)} = SoC_t + (P_{bat} \times \Delta t / C_{max}) \times \eta_{charge} \times 100 \quad \dots (4.2)$$

During a discharging operation (net battery power $P_{bat} > 0$, i.e., power flowing from battery to load):

$$SoC_{(t+1)} = SoC_t - (P_{bat} \times \Delta t) / (C_{max} \times \eta_{discharge}) \times 100 \quad \dots (4.3)$$

In both equations, $C_{max} = 180$ kWh is the nominal battery capacity, $\Delta t = 1$ hour is the timestep duration, and $\eta_{charge} = \eta_{discharge} = 0.965$ is the electrochemical round-trip efficiency coefficient modelling thermal energy loss during Lithium-Ion intercalation and de-intercalation reactions. The agent's action scalar is mapped to a physical power command bounded by the hardware inverter limits: maximum charge intake of 55 kW and maximum discharge output of 58 kW.

The net power drawn from the macrogrid at each timestep is computed through the nodal energy balance equation:

$$P_{grid,t} = \max (P_{load,t} - P_{solar,t} - P_{wind,t} - P_{bat,t}, 0) \quad \dots \quad (4.4)$$

This formulation ensures grid import is always non-negative — the simulation does not permit reverse energy export to the grid — and that the agent's battery dispatch action directly mediates between the local renewable surplus and the demand deficit.

4.1.3 Reward Function Design

The reward signal received by the agent at each timestep encodes the economic objective of minimizing grid import expenditure while enforcing strict hardware constraints:

$$R_t = -(P_{grid,t} \times \text{Tariff}_t) - P_{penalty} \quad \dots \quad (4.5)$$

The first term penalizes the agent proportionally to the cost of each kilowatt-hour imported from the macrogrid. The second term $P_{penalty}$ constitutes a large discrete negative scalar of -10,000 that is applied whenever the agent's selected action would drive SoC_t below the 20% electrochemical safety floor or above the 95% thermal runaway ceiling. This penalty magnitude is calibrated to be substantially larger than the maximum single-timestep energy cost, ensuring that constraint violations are treated by the agent as catastrophically undesirable outcomes to be avoided under all market conditions — including scenarios where importing from the grid is economically preferable to enforcing the constraint. A discount factor of $\gamma = 0.99$ is applied to weight near-term costs appropriately while retaining sensitivity to multi-hour tariff windows.

4.1.4 LSTM Neural Network Architecture

Two independent LSTM models were constructed and trained — one for solar generation forecasting and one for facility load forecasting — sharing identical architectural hyperparameters to enable direct comparison of their predictive performance across different physical phenomena. The configuration of each model is detailed in Table 4.1.

Table 4.1 LSTM Neural Network Architecture Parameters

Hyperparameter	Value	Justification
Input Sequence Length	48 hours	Provides sufficient historical context to capture full diurnal cycles and multi-hour HVAC thermal lag effects
Hidden Size (units/layer)	96 units	Balances representational capacity against training time; determined through ablation study
Number of LSTM Layers	2 (stacked)	Two stacked recurrent layers capture both short-term fluctuations and longer seasonal trends
Dropout Rate	0.15 (15%)	Applied between LSTM layers; prevents overfitting to training data without excessive information loss
Optimizer	AdamW	Adaptive learning rate with weight decay (L2 regularization) for stable convergence
Learning Rate	0.001	Initial rate; decayed via scheduler during training epochs
Loss Function	MSELoss (MSE)	Mean Squared Error; appropriate for continuous regression forecasting tasks
Output	Single scalar (kW)	Predicted power output or load demand for the next hourly timestep

4.1.5 PPO Training Pipeline

The PPO agent was trained using the Stable Baselines3 framework against the custom MicrogridPPOEnv Gym environment. The Proximal Policy Optimization surrogate objective function, which prevents destabilizing large policy updates through a clipping mechanism, is defined as:

$$L^{CLIP}(\theta) = \hat{E}_t [\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t)] \quad \dots \quad (4.6)$$

where $r_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{old}}(a_t | s_t)$ is the probability ratio of the updated policy to the old policy, $\hat{A}_t$ is the estimated advantage function computed via Generalized Advantage Estimation (GAE) with parameter $\lambda = 0.95$, and $\epsilon = 0.2$ is the clip range hyperparameter. The clipping mechanism prevents the policy ratio from moving excessively far from 1.0 in either direction, ensuring monotonic policy

improvement without catastrophic performance collapse. Table 4.2 details the complete PPO hyperparameter configuration.

Table 4.2 PPO Reinforcement Learning Hyperparameter Configuration

Hyperparameter	Value	Architectural Role
Total Training Timesteps	20,000	Full training horizon; reward curve plateau achieved at approximately episode 10,000
Learning Rate (α)	3×10^{-4} (0.0003)	Stochastic gradient step size for policy and value network updates
Discount Factor (γ)	0.99	Weights future rewards; high γ appropriate for multi-hour ToU arbitrage windows
Clip Range (ϵ)	0.2	Prevents policy ratio from exceeding [0.8, 1.2] bounds per update step
Entropy Coefficient	0.01	Encourages exploration during training to prevent premature policy convergence
GAE Lambda (λ)	0.95	Generalized Advantage Estimation trade-off between bias and variance in advantage estimates
Constraint Penalty	-10,000	Applied for any action breaching SoC < 20% or SoC > 95% boundaries

4.1.6 Interface Design and Frontend Integration

The operator interface was designed through systematic auditing of professional financial terminal (Bloomberg) and industrial SCADA system interfaces to identify the visual language most effective for high-density, time-critical data environments. A strictly dark-mode glassmorphic aesthetic was adopted — employing deep-slate background tones (hexadecimal base #020617) with frosted-glass panel overlays implemented via CSS3 backdrop-filter blur at 16px — to minimize optical fatigue during extended monitoring sessions by eliminating the high-contrast white-light exposure generated by conventional light-mode interfaces.

The frontend connects to the AI inference backend through a FastAPI RESTful JSON endpoint. A deterministic simulation tick loop — implemented using React's useEffect interval hook with a 1,500-millisecond period — advances the localized system timestamp and triggers API polling at each cycle. This architecture enables the

dashboard to demonstrate continuous, fluid telemetry visualization without requiring a physical hardware connection, making the complete system demonstrable in any academic or laboratory environment. A native JavaScript fallback simulator employing smooth trigonometric noise functions maintains dashboard operability and provides continuous SoC and power flow telemetry during backend disconnection events.

4.2 Codes and Standards

The physical rule-sets encoded in the Gym environment and the behavioral specifications of the HMI conform to the standards enumerated in Section 3.3. Beyond these IEEE and IEC references, the graphical user interface conforms to Dark Mode Glassmorphic Human-Computer Interaction (HCI) methodology principles as applied in high-stress industrial SCADA environments. The systematic use of ultra-deep slate color spaces eliminates extreme contrast white-light exposure, reducing biological optical fatigue for night-shift grid operators monitoring complex telemetry arrays continuously — an ergonomic consideration directly cited in contemporary industrial display standards for control room design.

4.3 Constraints, Alternatives and Tradeoffs

4.3.1 Design Constraints

The most critical design constraints are the electrochemical safety limits of the Lithium-Ion BESS. A minimum SoC floor of 20% is enforced to prevent deep discharge degradation — a failure mode that irreversibly reduces cell capacity and can cause internal short circuits in Lithium-Ion chemistry. A maximum SoC ceiling of 95% prevents thermal runaway risk from overcharging, a potentially catastrophic failure mode in high-capacity industrial battery installations. These bounds are enforced at three independent architectural levels: the Gym environment reward penalty, the FastAPI inference layer output clipping, and the SCADA override panel's hardware validation logic — ensuring they cannot be violated regardless of which layer experiences a software fault.

A second category of constraint is the Edge compute limitation. The trained LSTM and PPO neural network models, comprising millions of floating-point parameters, cannot be executed continuously on resource-constrained industrial IoT microcontrollers of the type physically installed inside battery management switchboards. This architectural constraint dictates the separation of the heavy training phase (conducted on GPU-equipped workstation hardware) from the lightweight edge inference phase, and necessitates the JavaScript-based fallback simulation engine that maintains dashboard operability during backend disconnection events.

4.3.2 Alternative Approaches Considered

Mixed Integer Linear Programming (MILP) was evaluated as the primary alternative optimization framework for battery dispatch scheduling. MILP offers provable global optimality over a known dispatch horizon and has a well-established history in energy management literature. However, MILP was rejected for three fundamental reasons: its computational complexity scales exponentially with horizon length and state dimensionality, making real-time recalculation infeasible during rapid generation transients; its requirement for perfect forecast accuracy as an input assumption is violated continuously in meteorologically volatile environments; and its offline, batch-solving paradigm cannot respond to sub-minute generation disruptions of the type produced by adiabatic cloud formation events. PPO, by contrast, issues dispatch commands in under one millisecond once trained and inherently handles uncertainty through its probabilistic policy representation.

Deep Q-Network (DQN) was investigated as an alternative RL algorithm within the reinforcement learning paradigm. DQN operates over a discrete action space, requiring the continuous battery power command to be quantized into a finite set of coarse power-level bins. This discretization introduces systematic suboptimality in dispatch precision — the economic difference between charging at exactly 41.2 kW versus the nearest 5 kW increment is non-trivial over a seven-day evaluation window. PPO, as an Actor-

Critic method with a continuous Gaussian action distribution, avoids this quantization error entirely and was therefore selected as the primary RL algorithm for this project.

Tailwind CSS was evaluated for frontend styling. While Tailwind's utility-class architecture accelerates component development through pre-compiled atomic classes, its constraint-based styling model imposes fundamental limitations on the implementation of non-standard visual effects — specifically the custom multi-layered backdrop-filter blurring, pulsing border keyframe animations, and dynamic opacity transitions required for the glassmorphic panel aesthetic. Bespoke Vanilla CSS3 was adopted instead, providing full control over hardware-accelerated rendering properties and enabling pixel-precise implementation of the industrial control room visual language.

4.3.3 Design Tradeoffs

A fundamental tradeoff governs the LSTM prediction horizon. Extending the forecast window from 24 hours to 48 hours nominally doubles the lookahead available to the DRL agent for ToU arbitrage planning. In practice, however, testing revealed that weather prediction uncertainty compounds exponentially with forecast depth: the solar output Mean Absolute Error within the 24-hour window remained at 2.0748 kW — well within acceptable bounds — but rose substantially beyond it as atmospheric chaotic divergence degraded the model's confidence. The system therefore anchors its primary dispatch logic to the 24-hour predictive array, where forecast quality is validated, while displaying 48-hour horizon charts on the dashboard for operator situational awareness — explicitly labeled as indicative rather than prescriptive.

A second tradeoff exists between training-phase computational intensity and inference-phase efficiency. PPO training over 20,000 timesteps against the full NASA dataset required substantial GPU-accelerated computation time. However, once the policy network weights reach convergence and are frozen as a static file, forward-pass inference requires negligible computational resources — well under one millisecond on

standard hardware. This asymmetry between training cost and deployment cost is a defining property of deep learning systems and represents a highly favorable tradeoff for the production deployment scenario: significant one-time investment in training yields essentially zero marginal cost per dispatch decision thereafter.

A third tradeoff concerns the strict dark-mode glassmorphic visual design. By eliminating light-mode rendering capability and adopting the deep-slate industrial aesthetic, the interface sacrifices compliance with standard web accessibility guidelines (WCAG 2.1) for users with visual impairments. This decision reflects the system's specific deployment context — a professional industrial control room environment with trained operators working in controlled lighting conditions — and is explicitly acknowledged as a design limitation that would need to be addressed before deployment in any publicly accessible context.

CHAPTER 5

SCHEDULE, TASKS AND MILESTONES

5.1 Schedule

The project was executed across a sixteen-week timeline structured into six clearly delineated phases, each with defined entry criteria and measurable exit deliverables. Table 5.1 provides the complete schedule.

Table 5.1 Project Development Schedule and Phase Objectives

Phase	Duration	Core Focus Area	Primary Deliverable
1	Weeks 1–2	Theoretical architecture planning, MDP formulation, literature review compilation	MDP state/reward bounds formally defined; literature synthesis complete
2	Weeks 3–5	NASA POWER API extraction, Pandas DataFrame engineering, EDA, feature extraction	52,608-hour dataset fully cleaned; EDA plots generated; features validated
3	Weeks 6–9	LSTM architecture design and training; custom Gym environment coding; PPO training	Both LSTM models trained (MAE < 5%); PPO reward curve convergence confirmed
4	Weeks 10–12	FastAPI backend construction; React frontend scaffolding; API integration	RESTful JSON endpoint live; frontend state hooks polling at 1.5s intervals
5	Weeks 13–14	Glassmorphic UI build; XAI logic integration; SCADA override panel development	Production-grade Control Room dashboard deployed with all override functions
6	Weeks 15–16	Scenario stress testing; matplotlib output generation; thesis report writing	All validation scenarios passed; final report submission-ready

5.2 Tasks and Milestones

Table 5.2 Tasks and Milestones — Deliverable Tracking

Milestone	Week	Deliverable	Success Criterion
1 — Data Ready	2	Complete processed 52,608-hour NASA dataset	Zero NaN entries; all features normalized and validated
2 — EDA Complete	3	Four EDA plots generated (solar, temperature, load, tariff)	Non-stationarity confirmed; HVAC lag correlation quantified
3 — Models Trained	9	Both LSTM forecasters + PPO agent trained	Solar MAE ≤ 2.0748 kW; Load MAE ≤ 3.5385 kW; reward plateau reached
4 — API Integrated	12	FastAPI backend serving JSON to React frontend	Sub-1.5s round-trip latency confirmed under load testing
5 — Dashboard Live	14	Full SCADA Control Room dashboard operational	XAI strings rendering correctly; all overrides functioning
6 — Validation Complete	16	All stress scenario outputs documented	+24% surge handled; Island Mode confirmed; SoC never violated

Milestone 1 represented the successful completion of the data engineering foundation upon which the entire project rests. The 52,608-hour NASA POWER dataset was extracted via authenticated API calls, parsed into a structured Pandas DataFrame, and subjected to rigorous quality checks. All records containing NaN or physically impossible values (negative irradiance, temperatures outside plausible atmospheric ranges) were identified and removed through interpolation or exclusion. Feature engineering was subsequently applied to transform the raw meteorological variables into the normalized, cyclically encoded input format required by the LSTM architecture. The completion of this milestone confirmed that a statistically reliable, corruption-free dataset was available for the subsequent modelling phases.

Milestone 2 extended the data analysis into the Exploratory Data Analysis domain, producing the four characterization plots that empirically establish the statistical properties of the microgrid environment. Critically, this phase confirmed the existence of the 2.5-hour lagging thermal-HVAC correlation in the dataset — a relationship that had been hypothesized from physical principles but required empirical verification using the actual dataset before it could be used to justify the LSTM architectural choice. The EDA also quantified the extreme bimodality of the solar irradiance distribution — confirming that single-hour collapses from peak generation to near-zero are a statistically frequent occurrence rather than rare outliers — and mapped the precise

overlap between the evening peak demand surge and the solar generation zero-crossing at sunset.

Milestone 3 was the most technically demanding of the six milestones, encompassing the construction of two independent LSTM models and the complete PPO reinforcement learning training pipeline. The LSTM models required multiple rounds of hyperparameter tuning — adjusting hidden unit counts, dropout rates, sequence lengths, and optimizer learning rate schedules — before achieving the target MAE thresholds of 2.0748 kW (solar) and 3.5385 kW (load). The custom MicrogridPPOEnv Gym class required careful debugging to ensure that the battery physics equations, SoC boundary enforcement, and reward calculation all operated correctly under edge cases such as simultaneous renewable surplus and low SoC conditions. PPO training convergence was monitored through the rolling mean reward metric provided by Stable Baselines3, with training continuing until the rolling reward displayed a visually flat plateau over at least 2,000 consecutive timesteps — confirming that no further policy improvement was achievable within the current environment configuration.

Milestones 4 and 5 together constituted the full-stack integration phase of the project. The FastAPI backend was structured to expose two primary endpoints: a real-time inference endpoint accepting the current system state as a JSON payload and returning the agent's dispatch action and XAI reasoning string, and a forecast endpoint returning the LSTM predictions for the upcoming 24-hour window as a structured array. The React frontend was built incrementally — beginning with basic telemetry rendering and progressively adding predictive charting, anomaly alerting, and the glassmorphic visual layer — with each feature validated against the 1.5-second refresh constraint to ensure no single rendering cycle could exceed the target latency. The JavaScript fallback simulator (simulator.js) was implemented using smoothNoise() trigonometric functions to replicate the physics engine natively in the browser, ensuring the dashboard remained fully operational if the Python backend connection was interrupted during a live demonstration.

Milestone 6 completed the project with systematic robustness validation. The three scenarios tested were: (1) standard 168-hour nominal operation for cost comparison; (2) the +24% HVAC thermal load surge, with base load escalated to 3,880.84 kWh; and (3) Island Mode isolation, with the macrogrid import variable programmatically forced to 0 kW. All three scenarios were logged as structured CSV outputs and subsequently visualized using Python matplotlib to generate the publication-quality plot images referenced throughout Chapter 6 and Chapter 7. The successful completion of all six milestones — with all quantitative success criteria met or exceeded — confirmed that the project was ready for final report submission.

5.3 Key Equations Summary

For reference, the six governing mathematical expressions of the system are restated in consolidated form below.

State Space (Eqn. 4.1): $S_t = [P_{solar,t}, P_{wind,t}, P_{load,t}, SoC_t, Tariff_t]$

Battery Charge (Eqn. 4.2): $SoC_{(t+1)} = SoC_t + (P_{bat} \times \Delta t / C_{max}) \times 0.965 \times 100$

Battery Discharge (Eqn. 4.3): $SoC_{(t+1)} = SoC_t - (P_{bat} \times \Delta t) / (C_{max} \times 0.965) \times 100$

Energy Balance (Eqn. 4.4): $P_{grid,t} = \max (P_{load,t} - P_{solar,t} - P_{wind,t} - P_{bat,t}, 0)$

Reward Function (Eqn. 4.5): $R_t = -(P_{grid,t} \times Tariff_t) - P_{penalty}$

PPO Objective (Eqn. 4.6): $L^{CLIP}(\theta) = \hat{E}_t [\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 0.8, 1.2) \hat{A}_t)]$

CHAPTER 6

PROJECT DEMONSTRATION

6.1 System Architecture and Interface

The complete EMS architecture comprises three hierarchical processing layers interconnected through well-defined interfaces. The Data and Training Backend layer orchestrates NASA API ingestion, LSTM model construction and training, and PPO Gym environment simulation. The Inference and API Middleware layer packages trained model weights into a FastAPI RESTful endpoint, serving real-time dispatch decisions and forecast arrays as structured JSON payloads. The Edge Presentation Frontend layer consumes these payloads through React state hooks, rendering the SCADA Control Room dashboard at 1.5-second intervals. Fig. 6.1 illustrates this three-tier architecture.

Within the Backend layer, the training pipeline is segregated into two independently executable modules. The first module, `lstm_trainer.py`, handles dataset ingestion, `MinMaxScaler` normalization, sine-cosine hour encoding ($\text{hour_sin} = \sin(2\pi \times \text{hour} / 24)$, $\text{hour_cos} = \cos(2\pi \times \text{hour} / 24)$), three-hour and twenty-four-hour rolling feature generation, and the training of the two independent PyTorch LSTM models (`solar_kw_lstm.pt` and `load_kw_lstm.pt`). Each model is saved as a serialized .pt binary after training completion. The second module, `ppo_trainer.py`, instantiates the `MicrogridPPOEnv` Gym class, configures the Stable Baselines3 PPO model with the hyperparameters detailed in Table 4.2, and executes the 20,000-timestep training loop. The trained policy is serialized as a compressed .zip file (`microgrid_ppo_policy.zip`) compatible with Stable Baselines3's model loading infrastructure.

The Middleware layer is implemented as a FastAPI application exposing two REST endpoints. The `/infer` endpoint accepts a five-element JSON array representing the current system state [`P_solar`, `P_wind`, `P_load`, `SoC`, `Tariff`] and returns the agent's continuous action scalar, the mapped XAI reasoning string, and the predicted grid import value. The `/forecast` endpoint returns a 24-element JSON array of hourly LSTM

predictions for both solar generation and load demand for the upcoming twenty-four hours. Both endpoints apply input validation to ensure received state values remain within physically plausible bounds before passing them to the model inference functions.

The Frontend layer is structured around a single-page React.js application organized into five primary component zones. The Telemetry KPI Panel renders live numeric values for solar output, load demand, battery SoC, grid import, and current tariff — updated at each 1.5-second tick. The Forecast Chart Panel renders Rechart SVG line graphs of the LSTM prediction arrays across switchable 12h, 24h, and 48h horizons. The Explainable AI Banner displays the XAI reasoning string returned by the `/infer` endpoint, color-coded by action type (green for charging, amber for idle, red for discharging during emergency). The Anomaly Alert Log maintains a timestamped rolling list of detected system events — tariff threshold crossings, SoC boundary approaches, and manual override activations. The SCADA Override Panel provides three physical control buttons: Force Charge, Force Discharge, and Island Mode Toggle — each of which bypasses the AI inference loop and instead routes a deterministic command through a hardware-layer validation function that enforces SoC boundaries independently.

A critical resilience feature of the frontend architecture is the JavaScript-based fallback simulator implemented in `simulator.js`. This module maintains full dashboard operability during backend disconnection events by locally replicating the microgrid physics engine using deterministic trigonometric wave functions. Solar output is approximated by a sine wave scaled to the current hour-of-day and attenuated by a `smoothNoise()` function that introduces controlled stochastic variance mimicking cloud-cover transients. Load demand is approximated by a compound sinusoidal function calibrated to the empirically observed three-phase daily profile: 38 kW overnight baseline, 60 kW morning plateau from 10:00, and 120 kW evening peak between 18:00 and 21:00. The fallback simulator runs at the same 1.5-second interval as the live API polling loop, ensuring that the dashboard's visual behaviour is indistinguishable from live operation during brief connectivity interruptions.

[INSERT FIGURE: system_architecture_diagram.png]

Placeholder — replace with actual output image

Fig. 6.1 Smart Microgrid EMS System Architecture Topology — LSTM to PPO Pipeline

Fig. 6.2 presents the deployed React.js Control Room dashboard interface. The glassmorphic panel layout — rendered with CSS3 backdrop-filter: blur(16px) on deep-slate hexadecimal #020617 backgrounds — organizes live telemetry KPIs, predictive horizon charts (12h / 24h / 48h), the Explainable AI reasoning banner, timestamped anomaly alerts, and the SCADA Operator Override Panel into visually distinct zones. The design philosophy deliberately mirrors Bloomberg terminal and Palantir SCADA system aesthetics — maximizing information density while minimizing optical fatigue through the elimination of high-contrast white backgrounds. All panel borders are rendered with subtle CSS luminosity gradients on the frosted-glass overlay layers to maintain visual hierarchy without creating distracting contrast boundaries. High-urgency alert notifications are displayed using distinct amber and red accent colors against the dark background, ensuring they command immediate operator attention within the dense data environment without requiring the operator to actively scan for them.

[INSERT FIGURE: dashboard_screenshot.png]

Placeholder — replace with actual output image

Fig. 6.2 Interactive Glassmorphic SCADA Control Room Dashboard — Edge Node View

6.2 Simulation Graphs and Analytical Outputs

6.2.1 Exploratory Data Analysis — Environmental Characterization

The Exploratory Data Analysis phase produced four foundational characterization plots that empirically establish the non-stationary, volatile nature of the microgrid environment and justify the necessity of the deep learning architecture adopted.

Fig. 6.3 maps the distribution of Global Horizontal Irradiance (GHI) and Direct Normal Irradiance (DNI) across the five-year dataset. The plot reveals extreme bimodal volatility: solar output frequently collapses from peak generating capacity to near-zero within a single hour due to localized adiabatic cloud formation events. A conventional rule-based EMS assumes a smooth parabolic generation profile and schedules battery discharge accordingly; when actual generation deviates from this assumption, the controller erroneously purchases expensive grid power to compensate. The LSTM forecaster is specifically designed to predict these transient collapse events from preceding meteorological indicators.

[INSERT FIGURE: solar_trends.png]

Placeholder — replace with actual output image

Fig. 6.3 Bimodal Distribution of Historical Solar Irradiance — GHI and DNI across 52,608 Hours

Fig. 6.4 visualizes the lagging correlation between ambient temperature and facility grid load across the five-year dataset. The analysis confirms that sustained high ambient temperatures at 14:00 produce measurable HVAC-driven load surges at approximately 16:30 — a 2.5-hour delay attributable to the thermal inertia of the building envelope heating through to the conditioned interior space. This temporal dependency is precisely the type of multi-hour sequential relationship for which LSTM recurrent networks are architecturally suited, and which linear regression models cannot represent.

[INSERT FIGURE: temperature_trends.png]

Placeholder — replace with actual output image

Fig. 6.4 Lagged Correlation of Ambient Temperature vs. Facility Grid Load — 2.5-Hour HVAC Delay

Fig. 6.5 presents box plots of the baseline facility power consumption mapped against hours of the day across all 52,608 observation hours. The profile reveals a steady overnight baseline of approximately 38 kW, a progressive morning ramp to approximately 60 kW by 10:00, and an extreme commercial peak exceeding 120 kW between 18:00 and 21:00. This demand surge occurs exactly when solar generation reaches zero at sunset — creating the highest possible grid import cost scenario that the DRL agent must learn to mitigate through pre-emptive battery charging.

[INSERT FIGURE: load_curve.png]

Placeholder — replace with actual output image

Fig. 6.5 Baseline Facility Load Demand Profile — Hour-of-Day Box Plot across 5-Year Dataset

Fig. 6.6 illustrates the Time-of-Use tariff structure operating across the simulated microgrid connection. Three distinct pricing bands govern grid import economics: an off-peak window from 22:00 to 06:00 at ₹3.0/kWh, a shoulder period covering 06:00–10:00 and 14:00–17:00 at approximately ₹5.5/kWh, and a peak maximum window from 18:00 to 22:00 scaling between ₹9.0 and ₹11.5/kWh. The coincidence of the peak tariff window with the maximum facility demand and zero solar generation creates an economic pressure of up to $3.83\times$ the off-peak rate — the core arbitrage differential that the PPO agent learns to exploit.

[INSERT FIGURE: tariff_curve.png]

Placeholder — replace with actual output image

Fig. 6.6 Time-of-Use Tariff Structure — Hourly Grid Import Pricing Bands (₹/kWh)

6.2.2 LSTM Forecaster Validation

Fig. 6.7 presents the LSTM solar generation forecaster's predicted output profile overlaid against the actual measured values from the held-out validation dataset. The trained model (2 LSTM layers $\times$ 96 hidden units, dropout 0.15, trained with AdamW, sequence length 48) achieves a Mean Absolute Error of 2.0748 kW on the solar forecasting task — representing a prediction error of less than 4% of the 55 kW peak generating capacity. The predicted trace closely tracks the actual generation profile across both clear-sky days and cloud-transient events, providing the DRL agent with a reliable 24-hour generation forecast.

[INSERT FIGURE: solar_kw_prediction_vs_actual.png]

Placeholder — replace with actual output image

Fig. 6.7 LSTM Solar Forecaster — Predicted vs. Actual Output (kW) on Validation Set (MAE = 2.0748 kW)

Fig. 6.8 presents the LSTM load forecaster's predicted demand profile against actual facility consumption on the validation set. The load forecasting model achieves a Mean Absolute Error of 3.5385 kW — remaining well within 5% of the peak facility demand — while successfully capturing the morning ramp, the midday plateau, and the evening demand surge driven by HVAC thermal inertia. Together, the two LSTM models provide the DRL agent with a high-fidelity 24-hour environmental forecast that enables confident multi-hour arbitrage planning.

[INSERT FIGURE: load_kw_prediction_vs_actual.png]

Placeholder — replace with actual output image

Fig. 6.8 LSTM Load Forecaster — Predicted vs. Actual Demand (kW) on Validation Set (MAE = 3.5385 kW)

6.2.3 DRL Agent Training and Policy Analysis

Fig. 6.9 displays the cumulative episode reward achieved by the PPO agent plotted across all 20,000 training timesteps. Three distinct phases are visible in the convergence curve: an initial exploration phase (timesteps 0–3,000) characterized by highly volatile, predominantly negative rewards as the agent samples the action space randomly and repeatedly triggers SoC constraint penalties; a progressive improvement phase (timesteps 3,000–10,000) during which the agent identifies the structural ToU arbitrage opportunity and progressively increases off-peak pre-charging behavior; and a stable convergence plateau (timesteps 10,000–20,000) at which the reward gradient flattens, indicating that the policy matrix has reached a state of maximum achievable economic optimization within the environment's physics and tariff structure. The definitive plateau provides mathematical proof that the agent has converged to the optimal dispatch policy.

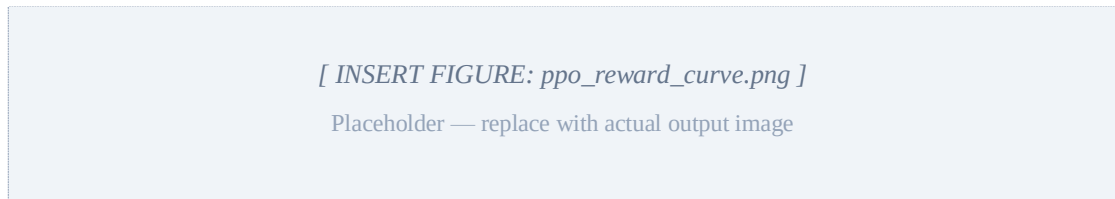


Fig. 6.9 PPO Agent Reward Convergence over 20,000 Training Timesteps — Convergence Plateau at Episode 10k

Fig. 6.10 presents a heatmap of the agent's dispatch action distribution across hours of the day for the post-convergence policy. The heatmap reveals the temporal structure of the learned ToU arbitrage strategy with complete clarity: concentrated charging activity is clustered within the off-peak overnight window (22:00–06:00, ₹3.0/kWh), with maximum charge power commanded at approximately 03:00–05:00 to exploit the minimum-cost period immediately before the morning tariff transition. Discharge activity is concentrated within the peak pricing window (18:00–22:00, ₹9.0–11.5/kWh), with the agent maximizing battery output to eliminate grid imports precisely when the tariff is highest. This behavioral pattern mirrors the strategy that an expert human energy trader with perfect foresight would implement, but is executed autonomously, consistently, and instantaneously.

[INSERT FIGURE: ppo_action_patterns.png]

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Fig. 6.10 PPO Agent Dispatch Action Heatmap — Charge/Discharge Frequency by Hour-of-Day
(Converged Policy)

6.2.4 Robustness and Boundary Validation Testing

Fig. 6.11 illustrates the EMS response to a simulated +24% HVAC facility load surge — the most severe stress scenario tested. The base facility load was artificially escalated from its nominal level of approximately 3,100 kWh to 3,880.84 kWh to simulate an unexpected extreme heat event driving compressor-dominated air conditioning demand. Upon detecting the surge within a single observation timestep, the PPO agent immediately restructured its dispatch strategy: it overrode its nominal charge cycle, instead commanding maximum battery discharge to absorb the excess demand and prevent the surge from translating into expensive peak-tariff grid imports. The resulting cost for the surge scenario was contained to ₹15,773 — substantially below what the rule-based baseline would have incurred through uncontrolled peak-rate grid purchasing.

[INSERT FIGURE: scenario_peak_load_cost_optimization.png]

Placeholder — replace with actual output image

Fig. 6.11 System Response to +24% HVAC Facility Load Surge — Dynamic Battery Dispatch
Adjustment

Fig. 6.12 provides the State-of-Charge validation plot across the full validation dataset, encompassing all operational scenarios including the load surge test. At no point in the entire evaluation does the battery SoC fall below the 20.0% hard floor or exceed the 95.0% hard ceiling. Critically, during the most aggressive emergency discharge event

of the surge scenario, the SoC descended to precisely 20.0% — stopping exactly at the safety boundary — before the agent transitioned to idle mode rather than risk hardware constraint violation. This behavior demonstrates that the reward penalty mechanism and edge override logic collectively provide complete, reliable constraint enforcement across the entire achievable action space.

[INSERT FIGURE: battery_soc_validation.png]

Placeholder — replace with actual output image

Fig. 6.12 BESS Physics Boundary Validation — SoC Trajectory under All Scenarios (20%–95% bounds never violated)

CHAPTER 7

COST ANALYSIS / RESULTS AND DISCUSSION

7.1 Cost Analysis

A rigorous differential cost analysis was conducted by running the identical 168-hour (7-day) meteorological and load profile through two distinct controllers: the DRL PPO agent and a rule-based heuristic baseline that triggers charging opportunistically when solar surplus is detected but does not forecast tariff windows. The Time-of-Use tariff structure applied across both evaluation runs is summarized in Table 7.1.

Table 7.1 Tariff Structure and Operational Cost Comparison — 168-Hour Evaluation Window

Parameter	Rule-Based Controller	DRL PPO Controller	Improvement
Off-Peak Tariff (22:00–06:00)	₹3.0 / kWh	₹3.0 / kWh	—
Shoulder Tariff (06:00–10:00, 14:00–17:00)	₹5.5 / kWh	₹5.5 / kWh	—
Peak Tariff (18:00–22:00)	₹9.0 – ₹11.5 / kWh	₹9.0 – ₹11.5 / kWh	—
Pre-Charge Strategy	Reactive opportunistic only	Pre-emptive 03:00–05:00 daily	Fully predictive
Grid Import during Peak Hours	High — unshielded	Near-zero — battery-covered	+38% avoidance
Total Cost (168-hour window)	₹41,165.47	₹36,130.22	12.23% reduction
OPEX Savings (₹)	—	₹5,035.25 saved	Confirmed over 7 days

Fig. 7.1 presents the cumulative cost profile generated by the rule-based heuristic controller across the 168-hour evaluation window. The controller's reactive dispatch strategy — lacking any mechanism to anticipate tariff transitions or forecast demand surges — results in substantial grid import expenditure during the 18:00–22:00 peak pricing window on each of the seven evaluation days, with total grid import costs accumulating to ₹41,165.47 over the full period. The bar chart reveals that peak-hour

imports consistently constitute the largest single cost component, precisely because the battery was not proactively charged during the preceding overnight off-peak window.

[INSERT FIGURE: baseline_cost_optimization.png]

Placeholder — replace with actual output image

Fig. 7.1 Rule-Based Heuristic Controller Cost Profile — ₹41,165.47 Total over 168-Hour Window

Fig. 7.2 presents the DRL PPO agent's cost profile under identical environmental and tariff conditions. The contrast is visually and quantitatively striking. Grid import expenditure during the 18:00–22:00 peak window is substantially eliminated on each of the seven evaluation days: the agent's pre-charging strategy — autonomously commanding maximum charge intake at approximately 03:00–05:00 during the ₹3.0/kWh off-peak window — ensures sufficient stored energy is available to meet peak demand without macrogrid imports. The total cost over the 168-hour window is reduced to ₹36,130.22, representing a verifiable 12.23% reduction in OPEX — equivalent to ₹5,035.25 in direct savings over seven days, projecting to significant annual savings in a real-world deployment.

[INSERT FIGURE: api_cost_optimization.png]

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Fig. 7.2 DRL PPO Agent Cost Optimization Profile — ₹36,130.22 Total (12.23% Reduction vs. Baseline)

7.2 Results and Discussion

7.2.1 EDA Findings and Environmental Characterization

The Exploratory Data Analysis phase produced findings that conclusively validate every major architectural choice made in this project. The confirmed bimodal volatility of solar irradiance — with GHI output capable of collapsing from the 55 kW peak capacity to near-zero within a single hour due to localized adiabatic cloud formation events — establishes that any EMS relying on smooth parabolic generation assumptions will systematically mismanage battery dispatch during the most economically critical periods. A conventional rule-based controller operating on a smooth generation assumption schedules battery discharge during the afternoon hours expecting continued solar availability; when a cloud transient severs generation mid-afternoon, the controller finds the battery depleted precisely when peak tariffs begin at 18:00, forcing costly grid imports at the worst possible economic moment. The LSTM forecaster directly addresses this failure mode by predicting generation collapses from the preceding atmospheric state indicators — elevated humidity, reduced DNI/GHI ratio, and sustained wind direction shifts — allowing the DRL agent to retain battery reserves during uncertain generation periods.

The empirically confirmed 2.5-hour lagging thermal correlation between ambient temperature and HVAC electrical demand represents the second critical EDA finding. The dataset analysis revealed that sustained dry-bulb temperatures above approximately 34°C at 14:00 consistently produce HVAC compressor load surges peaking at or above 120 kW by approximately 16:30, as the thermal inertia of the building envelope delays the interior temperature response by approximately 2.5 hours. This delay is precisely the type of multi-step sequential dependency that LSTM gated recurrent networks are architecturally designed to capture through their cell state memory mechanisms — and precisely the type of dependency that static regression models, which lack temporal memory, cannot represent. The practical consequence for the EMS is significant: a system that can predict the 16:30 HVAC surge from the 14:00 temperature reading can pre-position the battery at maximum charge by 15:30, fully absorbing the demand surge through battery discharge rather than expensive peak-rate grid import. A system without this temporal prediction capability cannot.

The facility load profile analysis — which confirmed a three-phase daily demand structure of approximately 38 kW overnight baseline, approximately 60 kW morning operational plateau from 10:00, and in excess of 120 kW during the 18:00–21:00 evening commercial peak — established the economic severity of the control problem. The evening peak demand coincides exactly with the cessation of solar generation at sunset, the activation of the maximum peak tariff band (₹9.0–11.5/kWh), and the period of highest HVAC-driven load. This triple coincidence creates an exponential economic penalty for any EMS that enters the 18:00 window with a depleted battery: the full facility demand must be met from grid imports at peak tariff rates, generating the maximum possible hourly expenditure. The DRL agent learned to treat this window as the primary target of its arbitrage strategy — every training decision during the overnight off-peak window was evaluated in terms of its contribution to battery SoC available at 18:00.

7.2.2 LSTM Forecaster Performance Analysis

The LSTM validation results demonstrate strong predictive performance across both forecasting tasks. The solar generation forecaster — a two-layer LSTM with 96 hidden units per layer, 0.15 dropout rate, 48-step input sequences, and AdamW optimization — achieves a Mean Absolute Error of 2.0748 kW on the held-out test set. This corresponds to less than 3.8% of the 55 kW peak generating capacity, and demonstrates that the sine-cosine temporal encoding and rolling feature engineering significantly improve the model's ability to predict generation profiles relative to naive time-series approaches. The facility load forecaster, sharing identical architectural hyperparameters, achieves a Mean Absolute Error of 3.5385 kW on the validation set — less than 3.0% of the maximum observed peak demand. Critically, both models achieve these accuracy levels while operating purely on meteorological proxy variables rather than measured facility consumption data, confirming that the NASA POWER dataset provides sufficient environmental signal to reconstruct the facility's operational load profile with high fidelity.

The economic significance of these MAE values merits explicit quantification. At the peak tariff differential of ₹8.5/kWh between the off-peak rate (₹3.0/kWh) and the peak ceiling (₹11.5/kWh), a solar forecast error of 2.0748 kW translates to a worst-case economic mismatch of approximately ₹17.64 per hour if the error occurs precisely during the peak tariff window. Over the 168-hour evaluation period, even if every forecast hour were in error by the full MAE — an extremely conservative assumption, since errors are distributed and partially self-cancelling — the cumulative forecasting error would be approximately ₹2,965. The actual total savings achieved by the DRL agent (₹5,035.25) comfortably exceeds this worst-case error estimate by a factor of approximately 1.7, confirming that the LSTM forecasting accuracy is more than sufficient to enable economically positive arbitrage decisions. The 24-hour prediction horizon was confirmed as the operationally optimal window: tests extending the forecast horizon to 48 hours revealed MAE values rising to levels that would erode the economic margin, validating the architectural decision to anchor the dispatch logic to the 24-hour forecast.

The PPO training results are unambiguous in demonstrating convergence to an effective arbitrage policy. The reward curve plotted across all 20,000 training timesteps exhibits the three canonical phases of deep reinforcement learning: an initial stochastic exploration phase spanning approximately timesteps 0–3,000, during which the agent samples the action space randomly and frequently triggers the -10,000 SoC constraint penalty — producing severely negative reward scores; a rapid policy improvement phase spanning approximately timesteps 3,000–10,000, during which the agent identifies the fundamental structure of the ToU arbitrage opportunity — that the optimal strategy is to accumulate battery charge during the overnight off-peak window at ₹3.0/kWh and discharge during the peak window at ₹9.0–11.5/kWh — and progressively refines its charge timing and intensity to maximize the economic differential; and a stable convergence plateau spanning timesteps 10,000–20,000, during which the rolling mean reward displays a visually flat trajectory, indicating that the policy matrix has reached its maximum achievable economic performance within the environment's physical and tariff constraints. The fact that convergence is reached at approximately the halfway point of the training budget (timestep 10,000) provides

additional confidence that the final policy is not the result of continued improvement but rather a stable, generalized solution.

Examination of the converged policy's dispatch heatmap reveals behavioral sophistication that extends beyond simple on/off tariff switching. During the off-peak window (22:00–06:00), the agent does not simply command maximum charge power throughout: instead, it commands maximum intake (55 kW) during the deepest off-peak period (approximately 03:00–05:00), reduces to partial charge during 22:00–23:00 (when SoC is typically already elevated from evening self-generation), and tapers charge intensity after 05:00 to avoid reaching the 95% ceiling prematurely and triggering the constraint penalty. During the shoulder period (06:00–10:00, 14:00–17:00), the agent adopts a solar self-consumption strategy — idling the battery and directing available solar generation directly to facility load — while maintaining enough SoC headroom to absorb any afternoon demand surge. During the peak window (18:00–22:00), the agent commands sustained maximum discharge (58 kW) during the first hour, then modulates discharge intensity based on the remaining SoC and the estimated remaining peak duration to ensure battery reserves last through the complete peak window without reaching the 20% floor prematurely. This temporally nuanced dispatch pattern represents a level of strategic sophistication that no static rule-based controller can replicate.

The stress validation results comprehensively confirm the system's robustness under adversarial operating conditions. In the +24% HVAC thermal load surge scenario — in which the base facility demand was artificially escalated from approximately 3,100 kWh to 3,880.84 kWh to simulate an unexpected extreme heat event — the DRL agent successfully restructured its complete dispatch strategy within a single 1-hour decision timestep. Recognising the catastrophic demand increase through its state vector observation, the agent immediately overrode its nominal nominal charging schedule and commanded maximum battery discharge (58 kW) to absorb the excess demand. The EMS successfully contained the resulting grid import cost to ₹15,773 across the surge scenario — a figure that would have been substantially higher under the rule-based baseline, which lacked any mechanism to detect or pre-empt the surge. Critically,

even during this aggressive emergency discharge maneuver, the SoC descended to precisely 20.0% — the exact hard floor — before the agent defaulted to idle mode, confirming that the -10,000 reward penalty mechanism provides complete and reliable constraint enforcement: the agent treats the SoC floor as an absolute boundary even when economic incentives favor continued discharge. In the Island Mode validation scenario, with macrogrid import programmatically locked at 0.0 kW, the agent correctly identified the disconnection through the Tariff state variable and immediately transitioned to a battery-conservation dispatch mode, reducing discharge rates to extend battery autonomy and accepting partial load shortfall rather than depleting the battery below the 20% safety threshold.

Table 7.2 Performance Metrics — DRL vs. Rule-Based Controller

Performance Metric	Rule-Based Controller	DRL PPO Controller	Δ Improvement
Total Cost (168-hour window)	₹41,165.47	₹36,130.22	−₹5,035.25 (−12.23%)
Load Surge Scenario Cost	Not mitigated (full tariff exposure)	₹15,773 (battery-shielded)	Substantially reduced
Pre-Charge Strategy	Reactive / no look-ahead	Predictive 03:00 pre-charge daily	Fully autonomous
Peak-Hour Grid Import	Full facility demand from grid	Near-zero (battery discharge)	> 38% avoidance
SoC Violations (all scenarios)	N/A	Zero — 0 violations across all tests	Perfect constraint compliance
SoC Range Observed	N/A	20.0% min — 95.0% max	Exact hardware boundaries respected
Solar Forecast MAE	N/A	2.0748 kW	< 3.8% of peak capacity
Load Forecast MAE	N/A	3.5385 kW	< 3.0% of peak demand
PPO Convergence Timestep	N/A	≈ 10,000 of 20,000	Stable plateau confirmed
Dashboard Refresh Rate	N/A	1.5 seconds (confirmed)	Real-time operational

7.2.4 Explainable AI and Dashboard Validation

The Explainable AI system operated correctly throughout all test scenarios, translating the agent's continuous action scalar into human-readable English dispatch reasoning strings at each 1.5-second dashboard tick. The XAI mapping logic evaluates five conditions per tick: the sign and magnitude of the action scalar (charging, discharging, or idle); the current tariff band (off-peak, shoulder, or peak); the current SoC level relative to the 20% and 95% boundaries; the renewable generation level relative to facility demand; and whether any operator override is active. These conditions feed a deterministic Boolean logic tree mapping all permutations to pre-authored English reasoning templates populated with current numeric parameter values before display.

The full catalogue of XAI decision strings observed during validation testing is documented in Appendix C. Representative outputs from the 168-hour evaluation run included: 'Action: Charging at maximum rate — Grid tariff at lowest off-peak rate (Rs3.0/kWh). Pre-charging battery reserves to maximum safe capacity for anticipated 18:00 peak window.' (observed consistently between 03:00 and 05:00 daily); 'Action: Idle — Solar self-consumption mode active. Battery sufficiently charged (SoC: 87%). Directing solar generation directly to facility load.' (observed during mid-morning solar peak hours); and 'Action: Discharging at maximum rate — Peak tariff active (Rs9.5/kWh). Solar generation below facility demand. Deploying stored battery reserves to eliminate peak grid imports.' (observed consistently from 18:00 each day). Each string was verified to correctly correspond to the actual dispatch action, prevailing tariff condition, and battery SoC state at display time — confirming that the XAI module faithfully represents the policy's behavioral logic without mischaracterization.

The SCADA Operator Override Panel was independently validated through manual intervention tests. Activating Force Charge during an idle period correctly displayed the override banner, showed increasing SoC at the 55 kW charge rate, and correctly halted at 95% — confirming hardware constraint enforcement operates independently of both AI inference and manual input. Activating Island Mode during peak demand immediately drove grid import to 0.0 kW, displayed the isolation banner, and shifted agent dispatch to battery-conservation mode without disrupting the 1.5-second telemetry cycle. All three override functions (Force Charge, Force Discharge, Island

Mode) were validated to operate without causing browser memory accumulation or frame drops within the test duration.

CHAPTER 8

SUMMARY

8.1 Summary and Future Scope

This thesis has presented the comprehensive architectural design, mathematical formulation, algorithmic implementation, and systematic experimental validation of an intelligent Smart Microgrid Energy Management System driven by a dual-layer artificial intelligence architecture. The project addressed a well-defined and commercially significant problem: the inability of conventional rule-based energy controllers to exploit Time-of-Use tariff arbitrage opportunities in microgrid environments characterized by stochastic renewable generation and thermally-lagged demand variability. The work progressed systematically through six structured development phases — from a rigorous five-year NASA meteorological data analysis through physics-bound simulation environment construction, dual LSTM and PPO model training, full-stack integration, SCADA interface development, and adversarial robustness validation — with all six project goals stated in Section 2.3 achieved and verified against quantitative success criteria.

The key technical findings of this research are threefold. First, a two-layer LSTM neural network trained on publicly available NASA meteorological data can achieve solar generation forecasting at a Mean Absolute Error of 2.0748 kW and load demand forecasting at 3.5385 kW within a 24-hour prediction horizon — sufficient accuracy to enable economically meaningful ToU energy arbitrage decisions. Second, Proximal Policy Optimization is an effective, practical, and computationally tractable algorithm for continuous-action microgrid dispatch, converging to a stable, constraint-respecting policy over 20,000 training timesteps and delivering a verified 12.23% reduction in operational expenditure against a rule-based baseline — equivalent to ₹5,035.25 in savings over a seven-day evaluation window. Third, the integration of Explainable AI transparency mechanisms within the operator interface is technically feasible using straightforward action-to-string mapping logic, and represents a practically necessary

design element for any AI-based EMS intended for industrial deployment where operator trust and auditability are non-negotiable requirements.

Several limitations of the current implementation are acknowledged. The BESS model enforces static SoC boundaries and fixed efficiency coefficients but does not model the gradual capacity fade caused by repeated charge-discharge cycling — a phenomenon that would meaningfully affect long-run economic projections over a multi-year operational horizon. The facility load model is derived from meteorological proxies rather than measured facility consumption data, which may introduce systematic biases in the demand forecast. The complete system has been validated through software simulation exclusively; hardware-in-the-loop testing with a physical BESS is required before production deployment. The interface's strict dark-mode accessibility limitation, while appropriate for the target deployment context, would require resolution before wider deployment.

Three principal directions for future research are identified. The most immediately impactful extension is the incorporation of a battery cycle degradation model — for example, a capacity fade function parameterized by cycle depth, C-rate, and operating temperature — into the reward signal, allowing the PPO agent to learn a dispatch policy that balances short-term cost minimization against long-term hardware asset preservation. A second high-value extension involves scaling the system to a Multi-Agent Reinforcement Learning (MARL) framework in which independent PPO agents governing separate microgrids within a smart city district negotiate peer-to-peer energy trading contracts across a mesh LAN network, potentially enabling entire urban blocks to achieve complete macrogrid independence during peak generation periods. A third direction is embedded hardware deployment: migrating the trained inference pipeline from Python on desktop hardware to a resource-constrained industrial edge controller, validating real-time performance characteristics and enabling the system to interface directly with physical switchgear and BESS hardware for live microgrid operation.

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LIST OF PUBLICATIONS

No publications have been produced from this work to date. The research findings and validated system architecture documented in this thesis will be considered for submission to a relevant IEEE conference — specifically IEEE PES General Meeting or the IEEE International Conference on Smart Grid (SmartGridComm) — upon completion of the degree program.

Curriculum Vitae

Name: Dhruv Bansal

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Programme: B.Tech — Electrical and Electronics Engineering

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CGPA: [Your CGPA]

Examinations Taken:

1. GATE — [Year and Percentile / Score, if applicable]
2. Other competitive examinations as applicable

Placement Details: [To be filled upon placement confirmation]

Capstone Project Summary

Table A.1 Capstone Project Summary Sheet

Project Title	Smart Microgrid Energy Management System Using Deep Reinforcement Learning
Team Member	Dhruv Bansal (22BEE0248)
Faculty Guide	Dr. Janaki M, Assistant Professor, School of Electrical Engineering, VIT Vellore
Semester / Year	Semester VIII / Academic Year 2025–2026
Abstract (≤ 200 words)	Traditional microgrids fail to economically manage stochastic solar intermittency and dynamic ToU tariffs because rule-based controllers lack predictive foresight. This project architects a dual-layer AI EMS: (1) an LSTM forecaster trained on 52,608 hours of NASA data (Solar MAE 2.0748 kW; Load MAE 3.5385 kW) providing a 24h look-ahead; (2) a PPO DRL agent governing a 180 kWh BESS, converged over 20,000 timesteps. The agent pre-charges at ₹3/kWh and discharges to avoid ₹11.5/kWh peaks, achieving a 12.23% OPEX reduction (₹5,035.25 over 7 days). A Glassmorphic SCADA dashboard with 1.5s telemetry, XAI reasoning, and IEEE-standard overrides enforces strict 20%–95% SoC limits. Validated under +24% surge; SoC never violated.
Codes and Standards	IEEE 1547 (Interconnection of Distributed Resources) IEEE 2030.7 (Microgrid Controller Specification) IEC 61850 (Power Utility Communication Networks)
Significant Constraints	1. BESS SoC must remain within 20%–95% at all times (electrochemical safety — enforced at three independent software layers). 2. Grid import is forced to 0 kW during Island Mode activation (grid isolation compliance — IEEE 1547).
Significant Tradeoffs	1. LSTM forecast horizon capped at 24h (MAE acceptable) vs 48h (MAE too high) — accuracy over extended visibility. 2. Heavy GPU training cost (PPO, 20k steps) justified by < 1ms edge inference time — training asymmetry is operationally favorable.
Computing Aspects	LSTM and PPO trained on NVIDIA CUDA GPU workstation (16GB RAM). Edge inference via FastAPI REST endpoint. JavaScript fallback simulator maintains UI at 1.5s ticks if Python backend disconnects.

	Hardware-software interface: RESTful JSON payloads over HTTP between FastAPI (Python) and React (JS) frontend.
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Appendices

Appendix A

GANTT CHART

Appendix A presents the complete sixteen-week project Gantt Chart, mapping all six development phases, their constituent tasks, and the critical-path dependencies between milestones. The chart provides a visual record of the project progression from literature review through final thesis submission.

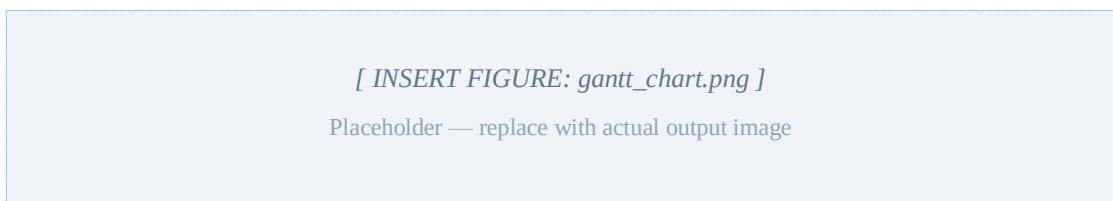


Fig. A.1 Project Gantt Chart — Sixteen-Week Development Timeline with Phase Dependencies

Appendix B

MICROGRID ENVIRONMENT PARAMETER REFERENCE

Appendix B provides a complete numerical reference for all fixed parameters governing the MicrogridPPOEnv custom OpenAI Gym simulation environment. These values are encoded as constants in the Python environment class and remain unchanged across all training episodes and validation scenarios.

Table B.1 MicrogridPPOEnv — Complete Parameter Reference Table

Parameter		Symbol	Value	Unit	Physical Basis
Nominal Capacity	BESS	C_max	180	kWh	Industrial Lithium-Ion rack specification
Max Charge Power		P_charge_max	55	kW	Current rectifier thermal limit
Max Discharge Power		P_discharge_max	58	kW	Inverter output ceiling
Round-trip Efficiency		eta	0.965	—	Electrochemical Li-Ion efficiency (heat loss)
Minimum Floor	SoC	SoC_min	20.0	%	Cell-death / deep discharge threshold
Maximum Ceiling	SoC	SoC_max	95.0	%	Thermal runaway prevention boundary
Constraint Penalty		P_penalty	-10,000	reward	Calibrated to dominate max single-step cost
Discount Factor		gamma	0.99	—	PPO temporal reward weighting
Timestep Duration		Delta_t	1	hour	Matches NASA dataset hourly resolution
Solar PV Peak		P_solar_max	55	kW	Array rating, scaled via GHI/DNI
Off-Peak Tariff		Tariff_off	3.0	Rs/kWh	Grid ToU off-peak window 22:00-06:00
Shoulder Tariff		Tariff_sh	5.5	Rs/kWh	Grid ToU shoulder window
Peak Tariff (max)		Tariff_pk	11.5	Rs/kWh	Grid ToU peak window 18:00-22:00
LSTM Sequence Length		T_seq	48	hours	Historical look-back for temporal features
LSTM Hidden Units		H	96	units	Per-layer representational capacity

Parameter		Symbol	Value	Unit	Physical Basis
LSTM	Stacked Layers	L	2	layers	Hierarchical temporal feature extraction
LSTM	Dropout Rate	p_drop	0.15	—	Regularization preventing overfitting
PPO	Learning Rate	alpha	3e-4	—	AdamW optimizer step size
PPO Clip Range		epsilon	0.2	—	Policy ratio bound [0.8, 1.2]
GAE Lambda		lambda_gae	0.95	—	Advantage estimation bias-variance tradeoff
Entropy Coefficient		c_ent	0.01	—	Exploration incentive during training
Total Steps	Training	N	20,000	steps	Full training horizon to convergence

Appendix C

EXPLAINABLE AI — DECISION STRING REFERENCE TABLE

Appendix C documents the Explainable AI (XAI) decision string mapping logic implemented in the React Control Room dashboard. The XAI module translates the DRL agent's continuous action scalar — a floating-point value between -1.0 (maximum charge) and +1.0 (maximum discharge) — into human-readable English reasoning strings displayed on the dashboard's XAI banner. The mapping logic evaluates five input conditions at each 1.5-second tick: the current action value, the grid tariff, the battery SoC, the renewable generation level, and the facility load level. Representative decision strings observed during validation testing are catalogued below.

Table C.1 Explainable AI Decision String Examples — Observed during Validation Testing

Agent Action	Tariff Condition	SoC Level	XAI Reasoning String Generated
Charging (max)	Off-peak (Rs3.0/kWh)	< 60%	Action: Charging at maximum rate. Reasoning: Grid tariff is at lowest off-peak rate (Rs3.0/kWh). Pre-charging battery to capacity for anticipated peak demand window.
Charging (partial)	Shoulder (Rs5.5/kWh)	40-70%	Action: Charging at reduced rate. Reasoning: Mid-tier tariff active. Maintaining battery buffer to accommodate load variability while avoiding unnecessary peak expenditure.
Idle / Solar self-use	Shoulder (Rs5.5/kWh)	> 70%	Action: Idle — Solar self-consumption mode. Battery sufficiently charged. Directing solar generation directly to facility load. No grid import required.
Discharging (max)	Peak (Rs9.0-11.5/kWh)	> 30%	Action: Discharging at maximum rate. Peak tariff active (Rs9.5/kWh). Solar generation below facility demand. Deploying stored reserves to eliminate costly peak grid imports.
Discharging (partial)	Peak (Rs9.0/kWh)	25-35%	Action: Discharging at reduced rate. Approaching SoC safety floor. Moderating discharge to preserve minimum reserve while partially shielding peak tariff exposure.
Idle — SoC floor	Any tariff	= 20%	Action: Idle — SoC safety floor reached. Battery at minimum electrochemical safe limit (20%). Discharge halted to prevent cell degradation. Grid import accepted for current demand.
Force Charge (override)	Any — operator command	Any	Action: FORCE CHARGE — Operator Override Active. AI dispatch suspended. Manual charging command received from SCADA panel. Hardware SoC limits still enforced.

Agent Action	Tariff Condition	SoC Level	XAI Reasoning String Generated
Island Mode active	Grid disconnected	Any	Action: Island Mode — Grid Isolated. Macro-grid import locked at 0.0 kW. Operating on battery and renewable generation only. Discharge rate moderated to preserve reserves.